Name: Garima Uppal CU240251618 sec:B roll no: 22

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.preprocessing import LabelEncoder, StandardScaler

print("liberaries are installed ") liberaries are installed

df =
pd.read_csv("/workspaces/Mastery_in_DataAnalytics_and_Visualizations_a
nd_Career_Advancemen/train-selected-columns.csv")

print("First 5 rows:")
print(df.head())

print("\nDataset Information:")
df.info()

print("\nStatistical Description:")
print(df.describe())
```

```
First 5 rows:
   PassengerId  Survived  Pclass  \
0            1         0       3
1            2         1       1
2            3         1       3
3            4         1       1
4            5         0       3

                                                Name     Sex   Age
SibSp  \
                                                                    0
                            Braund, Mr. Owen Harris    male  22.0   1
                                                                    1
  Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0   1
                                                                    2
                             Heikkinen, Miss. Laina  female  26.0   0
                                                                    3
       Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.0   1
                                                                    4
                           Allen, Mr. William Henry    male  35.0   0

   Parch            Ticket     Fare
0      0         A/5 21171   7.2500
1      0          PC 17599  71.2833


2      0  STON/O2. 3101282   7.9250
```

```
3      0            113803  53.1000
4      0            373450   8.0500

Dataset Information:
<class 'pandas.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 10 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
0   PassengerId  891 non-null    int64
1   Survived     891 non-null    int64
2   Pclass       891 non-null    int64
3   Name         891 non-null    str
4   Sex          891 non-null    str
5   Age          714 non-null    float64
6   SibSp        891 non-null    int64
7   Parch        891 non-null    int64
8   Ticket       891 non-null    str     9   Fare         891 non-
    null    float64 dtypes: float64(2), int64(5), str(3) memory
    usage: 69.7 KB

Statistical Description:
       PassengerId    Survived      Pclass         Age       SibSp  \
count    891.000000   891.000000   891.000000   714.000000   891.000000
mean     446.000000     0.383838     2.308642    29.699118     0.523008
std      257.353842     0.486592     0.836071    14.526497     1.102743
min        1.000000     0.000000     1.000000     0.420000     0.000000
25%      223.500000    0.000000    2.000000   20.125000    0.000000
50%      446.000000    0.000000    3.000000   28.000000    0.000000
75%      668.500000    1.000000    3.000000   38.000000    1.000000
max      891.000000    1.000000    3.000000   80.000000    8.000000

            Parch        Fare
count  891.000000  891.000000
mean     0.381594   32.204208  std
0.806057   49.693429  min
0.000000    0.000000  25%
0.000000    7.910400
50%      0.000000   14.454200  75%
0.000000   31.000000  max
6.000000  512.329200

print("Dataset Information:")
df.info()

Dataset Information:
<class 'pandas.DataFrame'>
RangeIndex: 891 entries, 0 to 890
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```
Data columns (total 10 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
0    PassengerId  891 non-null    int64
1    Survived     891 non-null    int64
2    Pclass       891 non-null    int64
3    Name         891 non-null    str
4    Sex          891 non-null    str
5    Age          714 non-null    float64
6    SibSp        891 non-null    int64
7    Parch        891 non-null    int64
8    Ticket       891 non-null    str     9   Fare         891 non-
null    float64 dtypes: float64(2), int64(5), str(3) memory usage:
69.7 KB
```

```
print("Statistical Description:")
df.describe()
```

```
Statistical
Description:
       PassengerId
Survived          Pclass
Age       SibSp  \
count         891.000000
891.000000    891.000000
714.000000    891.000000
mean          446.000000
0.383838        2.308642
29.699118       0.523008
std           257.353842
0.486592        0.836071
14.526497       1.102743
min             1.000000
0.000000        1.000000
0.420000    0.000000 25%
223.500000      0.000000
2.000000       20.125000
0.000000
50%           446.000000
0.000000        3.000000
28.000000       0.000000
75%           668.500000
1.000000        3.000000
38.000000       1.000000
max           891.000000
1.000000        3.000000
80.000000    8.000000
```

```
Parch       Fare
count
891.000000
891.000000  mean
0.381594
32.204208    std
0.806057
49.693429    min
0.000000
0.000000     25%
0.000000
7.910400
50%     0.000000
14.454200    75%
0.000000
31.000000    max
6.000000
512.329200
```

```
print("Dataset Shape:")
print(df.shape)
```

```
Dataset Shape:
(891, 10)
```

```
print("Missing Values:")
print(df.isnull().sum())
```

```
Missing Values:

PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
dtype: int64
```

```
print("Missing Value Percentage:")
print((df.isnull().sum() / len(df)) * 100)
```

```
Missing Value Percentage:
PassengerId     0.00000
Survived        0.00000
Pclass          0.00000
Name            0.00000
Sex             0.00000
Age            19.86532
SibSp           0.00000
Parch           0.00000
Ticket          0.00000
Fare            0.00000
dtype: float64
```

```
print("Duplicate records before removal:")
print(df.duplicated().sum())
df.drop_duplicates(inplace=True)

print("Duplicate records after removal:")
print(df.duplicated().sum())
```

```
Duplicate records before removal:
0 Duplicate records after
removal:
0
```

```
df =
pd.read_csv("/workspaces/Mastery_in_DataAnalytics_and_Visualizations_a
nd_Career_Advancemen/train-selected-columns.csv")
print(df.columns.tolist())
['PassengerId', 'Survived', 'Pclass', 'Name', 'Sex', 'Age', 'SibSp',
'Parch', 'Ticket', 'Fare']
```

```
df.to_csv("cleaned_train.csv", index=False)
print("Cleaned dataset saved successfully!")
Cleaned dataset saved successfully!
# Check duplicate records
print("Duplicate records before removal:")
print(df.duplicated().sum())

# Remove duplicate records
df.drop_duplicates(inplace=True)

# Check duplicates again
print("\nDuplicate records after removal:")
print(df.duplicated().sum())

Duplicate records before removal:
0

Duplicate records after removal:
0 # Step 7: Identify outliers using Box Plot and

IQR

import matplotlib.pyplot as plt
import seaborn as sns

# Box plots
plt.figure(figsize=(10, 5))
sns.boxplot(data=df[['Age', 'Fare']])
plt.title("Box Plot for Age and Fare")
plt.show()

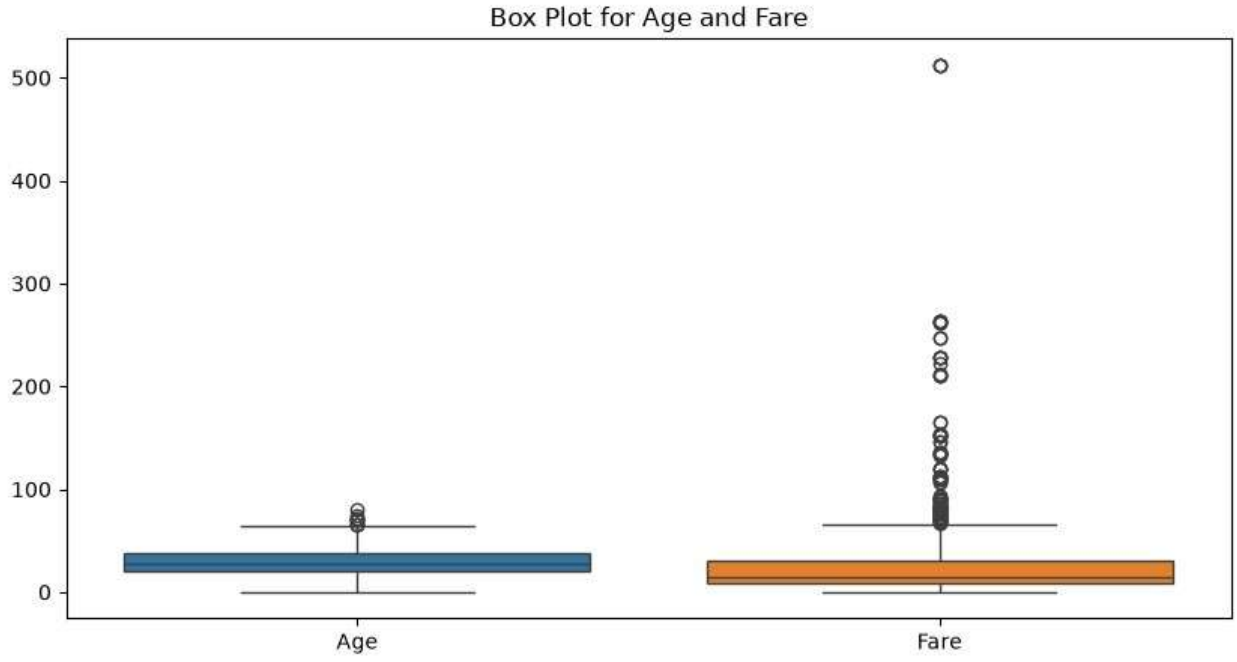
# IQR for Age
Q1 = df['Age'].quantile(0.25)
Q3 = df['Age'].quantile(0.75)

IQR = Q3 - Q1

lower_limit = Q1 - 1.5 * IQR
upper_limit = Q3 + 1.5 * IQR

# Remove Age outliers
df = df[(df['Age'].isna()) |
        ((df['Age'] >= lower_limit) & (df['Age'] <= upper_limit))]

print("Outliers in Age treated successfully.")
print("New dataset shape:", df.shape)
```


Box Plot for Age and Fare
500
400
300
200
100
0
Age
Fare

```
Outliers in Age treated successfully.
New dataset shape: (880, 10)

# Step 8: Normalization

from sklearn.preprocessing import MinMaxScaler

scaler = MinMaxScaler()
 Age and Fare normalized successfully.
c        Age      Fare                form(df[['Age', 'Fare']])

print("Age and Fare normalized successfully.")
print(df[['Age', 'Fare']].head())

 4  0.543882  0.015713                # Step 9: Feature Engineering

                                      0  0.339415  0.014151  # Create
                                      1  0.591066  0.139136  Family
Size                                  2  0.402328  0.015469
                                      3  0.543882  0.103644
                     df['FamilySize'] = df['SibSp'] + df['Parch'] + 1

# Create Age Group
df['AgeGroup'] = pd.cut(
df['Age'],
```

```
    bins=[0, 18, 35, 60, 100],
    labels=['Child', 'Young', 'Adult', 'Senior']
)

print("New features created successfully.")
print(df[['FamilySize', 'AgeGroup']].head())
New features created successfully.
   FamilySize AgeGroup
0           2    Child
1           2    Child
2           1    Child
3           2    Child
4           1    Child
# Step 10: Save the cleaned dataset

df.to_csv('titanic_cleaned.csv', index=False)

print("Cleaned Titanic dataset saved successfully!")

Cleaned Titanic dataset saved successfully!
```

QUESTION AND ANSWER

1. Why is data preprocessing important in data analytics?
Data preprocessing is the process of cleaning and transforming raw data before analysis. It is important because real-world data may contain missing values, errors, duplicates, and inconsistent formats. Proper preprocessing improves data quality, accuracy, and reliability of analysis and machine learning models.

2. Methods of handling missing values
Common methods include:

- **Removing records:** Delete rows containing missing values when only a few values are missing.
- **Mean:** Replace missing numerical values with the average.
- **Median:** Replace missing values with the median, especially when data contains outliers.
- **Mode:** Replace missing categorical values with the most frequent category.
- **Forward/Backward fill:** Use the previous or next available value, commonly for time-series data.

Example: If marks are 60, 70, missing, 80, the missing value can be replaced by the mean of the available marks.

3. Label Encoding vs One-Hot Encoding

Label Encoding	One-Hot Encoding
Assigns a number to each category	Creates separate columns for each category
Example: Male = 0, Female = 1	Male → [1,0], Female → [0,1]
Uses a single column	Creates multiple columns
Suitable for ordinal categories	Suitable for nominal categories

4. What are outliers and how do they affect results?

Outliers are observations that are significantly different from most other values in a dataset.

For example, if most salaries are ₹30,000–₹60,000 but one salary is ₹5,00,000, it may be an outlier.

Outliers can:

- Distort the mean.
- Affect statistical analysis.
- Influence regression models.
- Reduce machine learning performance.

5. Normalization vs Standardization

- **Normalization:** Scales values to a fixed range, usually **0 to 1.**
- **Standardization:** Transforms data so that it has a **mean of 0 and standard deviation of 1.**

Normalization is useful when features have different ranges, while standardization is commonly used for algorithms that depend on data distribution or distance.

6. Why should duplicate records be removed?

Duplicate records represent the **same observation more than once.** They can:

- Give incorrect counts.
- Bias statistical results.
- Increase the importance of repeated observations.
- Affect machine learning model training.

Removing duplicates ensures that each observation is represented correctly.

7. What is Feature Engineering?

Feature engineering is the process of creating, modifying, or selecting features from existing data to improve analysis or machine learning performance.

Examples:

1. Creating **Age** from an employee's date of birth.
2. Creating **Total Spending** by adding the prices of all purchases.

8. Preprocessing techniques for IBM HR Employee Attrition dataset

For the IBM HR Employee Attrition dataset, I would apply:

- **Missing-value handling** → To handle incomplete records.
- **Duplicate removal** → To avoid repeated employee records.
- **Categorical encoding** → Convert variables such as Department, Gender, and Job Role into numerical form.
- **Feature scaling** → Scale numerical features when required by the ML algorithm.
- **Outlier detection** → Identify unusual values in variables such as Monthly Income or Age.
- **Feature selection/engineering** → Select useful variables for predicting employee attrition.

These steps improve the **quality and performance** of the analysis and model.

9. How does poor-quality data affect machine learning?

Poor-quality data can lead to:

- Incorrect predictions.
- Lower model accuracy.
- Biased results.
- Overfitting or underfitting.
- Unreliable conclusions.

Therefore, **clean and properly prepared data** is essential for building effective machine learning models.

10. Three Python libraries for data preprocessing

Three commonly used Python libraries are:

1. **Pandas** – Data cleaning, missing values, filtering, and manipulation.
2. **NumPy** – Numerical operations and array processing.
3. **Scikit-learn** – Encoding, scaling, normalization, and preprocessing techniques.